

JEE MAIN REVISION MASTER: PROPERTIES OF TRIANGLE OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Properties Of Triangle — Core Principles & Governing Laws	27 PYQs	Core Principle
High	Properties Of Triangle — Standard Calculations & Problem Archetypes	9 PYQs	Quantitative Analysis
Standard	Properties Of Triangle — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Properties Of Triangle — Core Principles & Governing Laws	$\angle BAC = 90^\circ$, $\quad \theta_1 + \theta_2 = \frac{\pi}{2}$, $\quad c^2 = y$
Properties Of Triangle — Standard Calculations & Problem Archetypes	$\therefore A + C = \frac{2\pi}{3}$, $\quad 1, b =$, $\quad \frac{a+b}{7} = \frac{b+c}{8} = \frac{c+a}{9}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Properties Of Triangle — Core Principles & Governing Laws (Very High Yield — 27 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Properties Of Triangle.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Properties Of Triangle.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Properties Of Triangle PYQs.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "Let the area of a $\triangle PQR$ with vertices $P(5,4), Q(-2,4)$ and $R(a,b)$ be 35 square units. If its orthocenter and centroid are $O(2, \frac{14}{3})$..." [Ans: N/A]
- [2024 Evening Shift] Q#1 (Numerical): "Consider a triangle ABC having the vertices $A(1,2), B(\alpha, \beta)$ and $C(\gamma, \delta)$ and angles $\angle ABC = (\pi/6)$ and $\angle \dots$ " [Ans: 14]
- [2024 Morning Shift] Q#3 (MCQ): "Let $(5, (a/4))$ be the circumcenter of a triangle with vertices $A(a, -2), B(a, 6)$ and $C((a/4), -2)$. Let α denote the cir..." [Ans: N/A]

3.2 Properties Of Triangle — Standard Calculations & Problem Archetypes (High Yield — 9 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Properties Of Triangle.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2024 Evening Shift] Q#2 (MCQ): "Two vertices of a triangle ABC are $A(3, -1)$ and $B(-2, 3)$, and its orthocentre is $P(1, 1)$. If the coordinates of the point C are (x, y) " [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "For a triangle ABC, the value of $\cos 2A + \cos 2B + \cos 2C$ is least. If its inradius is 3 and incentre is M, then which of the following is NOT..." [Ans: N/A]
- [2022 Morning Shift] Q#4 (MCQ): "Let a, b and c be the length of sides of a triangle ABC such that $(a+b/7) = (b+c/8) = (c+a/9)$. If r and R are the radius of..." [Ans: N/A]

3.3 Properties Of Triangle — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Properties Of Triangle.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2025 Morning Shift] Q#1 (MCQ): "Let the area of a $\triangle PQR$ with vertices $P(5, 4)$, $Q(-2, 4)$ and $R(a, b)$ be 35 square units. If its orthocenter and centroid are $O(2, \frac{14}{5})$ and $C(c, d)$ respectively, then $c+2d$ is equal to" [Ans: N/A]
- [2024 Evening Shift] Q#1 (Numerical): "Consider a triangle ABC having the vertices $A(1, 2)$, $B(\alpha, \beta)$ and $C(\gamma, \delta)$ and angles $\angle ABC = (\pi/6)$ and $\angle ACB = (\pi/3)$ " [Ans: 14]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2025 (Online) 23rd January Morning Shift (Q#1)

Let the area of a $\triangle PQR$ with vertices $P(5, 4)$, $Q(-2, 4)$ and $R(a, b)$ be 35 square units. If its orthocenter and centroid are $O(2, \frac{14}{5})$ and $C(c, d)$ respectively, then $c+2d$ is equal to

- (A) 3
- (B) $(7/3)$
- (C) 2
- (D) $(8/3)$

Correct Answer: None

Solution Analysis: Equation of line QR = $5x + 2y + 2 = 0$ Equation of line PR = $10x - 3y - 38 = 0$ $\therefore$ Point R = $(2, -6)$ Centroid = $((5 + (-2) + a)/3, (4 + 4 + b)/3) = ((5 + (-2) + a)/3, (4 + 4 + b)/3)$ $c + 2d = (5/3) + (4/3) = 3$

Benchmark Problem 2: JEE Main 2024 (Online) 9th April Evening Shift (Q#2)

Two vertices of a triangle ABC are A(3,-1) and B(-2,3), and its orthocentre is P(1,1). If the coordinates of the point C are (α, β) and the centre of the circle circumscribing the triangle PAB is (h, k) , then the value of $(\alpha + \beta) + 2(h + k)$ equals

- (A) 81
- (B) 15
- (C) 51
- (D) 5

Correct Answer: None

Solution Analysis: $m_{PA} = \frac{2}{-2} = -1 \therefore m_{BC} = 1$ BC: $y = x + 5$ $m_{BP} = \frac{2}{-3} = -\frac{2}{3} \therefore m_{AC} = \frac{3}{2}$ AC: $y = \frac{3}{2}x - \frac{11}{2} \Rightarrow 2y = 3x - 11 \therefore C(21, 26)$ Let the circumcentre be (h, k) $(h-1)^2 + (k-1)^2 = (h+2)^2 + (k-3)^2 \dots$ (i) $(h-1)^2 + (k-1)^2 = (h-3)^2 + (k+1)^2 \dots$ (ii) Solving (i) and (ii) $\begin{aligned} & h = \frac{-19}{2}, k = \frac{-23}{2} \\ & \alpha + \beta + 2(h + k) = 21 + 26 - 19 - 23 = 2 + 3 = 5 \end{aligned}$